SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 30	Semester: July 2025-November 2025 Examination: In-Semester Examination (Mid-Semester Examination)	Duration: 1.15 Hrs

Programme code: 01 Programme: L.Y.BTECH	Class: LY	Semester: VII (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP
Course Code: 116U01E744	Name of the Course: Computer Simulation and Modeling	

Question No.		Max. Marks
Q1	Consider a single-server system. Let the arrival distribution be uniformly distributed between 1 and 8 minutes. Let the service distribution be as follows,	10

Service time (min)	1	2	3	4	5	6
Probability	0.17	0.15	0.32	0.20	0.06	0.10

Develop the simulation table and analyze the system by simulating the arrival and service of 10 customers. Assume that the first customer arrives at the system at the 0^{th} time. Random digits for inter-arrival time and service time are given below. Find server utilization, average waiting time, average service time, and average time spent by the customer in the system.

Customer	1	2	3	4	5	6	7	8	9	10
R D for Inter arrival	-	751	303	106	94	606	747	339	877	454
R D for Service time	74	52	16	82	94	61	87	35	29	99

Q2	A baker tries to determine how many dozens of bagels to bake every day. Probability distribution is as follows: -	10

No of customers/day	4	5	6	7
Probability	0.25	0.33	0.27	0.15

Customer buys 1, 2, 3 or 4 dozens bagels according to the following probability distribution.

No of dozens ordered /customers	1	2	3	4
Probability	0.20	0.35	0.15	0.30

Selling price of cake Rs. 5.00 per dozen. Cost to make per dozen is Rs. 3.00. End of the day the unsold bagels are sold at half the selling price. Assume that

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the baker baked 18 dozen every day, and **simulate for 2 days.**

R.D. for customer	**34**	**87**	**86**	**65**	**21**

R.D. for dozens ordered/customer: 18, 42, 34, 81, 61, 16, 30, 80, 21, 90, 69, 11, 40, 7, 55, 10.

OR

Explain the time advance algorithm. Tabulate clock system state, future event list, and cumulative statistics for a single-server queueing system with arrival and service details
Interarrival times: **3 2 6 2 4 5**
Service times: **2 5 5 8 4 5**
Stop the simulation when the clock reaches 20.

Q3 **Answer Any Two:** 10

i) Tom receives, on average, 6 emails a day (Poisson distributed). What is the probability that tomorrow's emails received will exceed the average by more than one standard deviation?

ii) The time to service customers at a bank teller is exponentially distributed with a mean of 50 seconds. What is the probability that the two customers in front of an arriving customer will each take less than 60 seconds to complete their transaction?

iii) The arrivals of customers at a teller counter follow a Poisson with a mean of 45 per hour, and the teller's service time follows an exponential with a mean of one minute. Determine steady state parameters LQ, L, WQ, and W.